Monte Carlo Simulation of the Black-Scholes price

Let $V(t)$ be the price of a simple option with payoff $\Phi(S)$ at maturity T for an underlying asset S following a geometric Brownian motion under martingale measure $\mathbb{Q}$ and r the risk-free interest rate. Then the terminal stock price can expressed (in distribution) as

$$S_T = S_T(Z) = S_0 \exp\left(\left(r - \frac{1}{2}\sigma^2\right) T + \sigma\sqrt{T} Z\right), \qquad Z \sim N(0,1), \tag{1}$$

which also gives a clean way to simulate the terminal stock price. The risk-neutral valuation formula gives

$$V(0) = e^{-rT}\mathbb{E}^{\mathbb{Q}}\left[\Phi(S_T)\right].$$

The plain Monte Carlo estimator of $V(0)$ is

$$\widehat{V}(0) = e^{-rT}\frac{1}{n}\sum_{i=1}^{n}\Phi(S_T(Z_i)), \qquad Z_i \overset{\text{i.i.d.}}{\sim} N(0,1). \tag{2}$$

To improve the variance of the estimator, consider the idea of antithetic variates, which is to pair each draw Z_i with $-Z_i$. Since the standard normal distribution is symmetric, so both $S_T(Z_i)$ and $S_T(-Z_i)$ have the correct risk-neutral Black–Scholes distribution. The improved antithetic estimator is therefore

$$\widehat{V}(0) = e^{-rT}\frac{1}{n}\sum_{i=1}^{n}\frac{\Phi(S_T(Z_i)) + \Phi(S_T(-Z_i))}{2}. \tag{3}$$

Note, if the payoff function Φ is increasing in S_T, then $S_T(Z_i)$ and $S_T(-Z_i)$ will tend to produce payoffs that are negatively correlated, which reduces the variance of the estimator. This is indeed the case for the European call and digital call options. The law of large numbers guarantees that (3) converges almost surely to the true price $V(0)$ as $n \to \infty$.